COMMUNITY TECH AID

SELF-HOSTED LLM · MULTI-DOMAIN INFRASTRUCTURE · PROJECT CASE STUDY

OPERATOR	ROLE	PLATFORM	STATUS
Kelly Guzman	**Founder & Sole Operator**	**communitytechaid.ai**	**Production**

Overview

Community Tech Aid (CTA) is a production technology platform operating under eBiz Market, LLC. I designed, built, and run it end-to-end: a self-hosted large language model behind a Cloudflare-fronted, multi-domain infrastructure. The full stack runs through me. Architecture, deployment, DNS, security, payments, email authentication, and day-to-day operations are all owned by a single operator.

This case study documents the technical scope of the platform and the security reasoning behind the architectural choices.

The self-hosted LLM

I built and deployed the LLM that powers `communitytechaid.ai` from the ground up, without prior formal training in machine learning operations. The decision to self-host rather than integrate a third-party API was deliberate and security-driven:

- **Data sovereignty.** User interactions never leave infrastructure I control. No third-party logging, no API-side data retention, no exposure to vendor policy changes.
- **Reduced supply-chain risk.** The platform does not depend on any single external vendor remaining accessible, affordable, or aligned with the platform's interests.
- **Predictable cost surface.** No per-token billing. Operational cost is fixed at the infrastructure layer rather than scaling unpredictably with traffic.
- **Privacy by architecture.** Self-hosting is the only configuration that allows hard guarantees about where data goes and who can access it.

The reasoning maps directly to core cybersecurity principles: data minimization, supply-chain risk reduction, and reducing the attack surface introduced by third-party dependencies.

Domain portfolio and DNS

I registered and manage several domains under the CTA and eBiz Market umbrella, centralized at Cloudflare for unified DNS, certificate, and security policy management. The primary user-facing domain is `communitytechaid.ai`. All variant domains redirect to the `.ai` primary, so any brand misroute (a user typing `.tech`, `.com`, etc.) still lands on the canonical platform without confusion or split traffic.

Cloudflare as proxy

All domains route through Cloudflare in proxy mode rather than DNS-only. The reasoning is security-first:

- **Origin IP concealment.** The actual server hosting the LLM never appears in public DNS. This shrinks the attack surface for direct-to-origin attacks, including reconnaissance and targeted exploit attempts.
- **DDoS mitigation.** Volumetric attacks are absorbed at Cloudflare's edge before they reach origin.
- **TLS termination and certificate lifecycle.** Edge handles HTTPS and certificate renewal for every domain, eliminating per-domain certificate juggling.
- **Web Application Firewall and bot management.** Edge-layer filtering of malicious requests, scrapers, and automated probes before they hit the application.
- **Caching and performance.** Static assets served from edge locations, reducing both latency and origin load.

Security operations

I own the full security posture of the platform:

- TLS configuration and certificate lifecycle management through Cloudflare
- DNS records, including SPF, DKIM, and DMARC for every sending domain
- Origin server hardening, patching cadence, and update discipline
- Access control and credential management for administrative interfaces
- Monitoring for anomalies in traffic and authentication patterns

Email infrastructure

Outbound email runs through Brevo, with SPF, DKIM, and DMARC records configured in Cloudflare DNS to authenticate the sending domains. This setup keeps marketing and

transactional email out of recipient spam folders and reduces the risk of the sending domains being abused for spoofing attacks against the platform's contacts.

Payments

Payment processing and reconciliation for the B2B service tier are managed directly by me, with the security, recordkeeping, and compliance considerations that come with handling customer financial data on a small-operator platform.

Community partners

The footer of `communitytechaid.ai` notes "With support from our community partners." This refers to local technology organizations and individuals in the North Idaho region who have provided guidance, referrals, or material support during the platform's build-out. The platform is not externally funded; it is independently designed, built, and operated.

Skills demonstrated

· Self-hosted LLM deployment, configuration, and operation

· Multi-domain DNS architecture with unified management

· Cloudflare proxy configuration including TLS, WAF, and DDoS mitigation

· Email authentication (SPF, DKIM, DMARC) across multiple domains

· Payment processing and customer financial data handling

· End-to-end security ownership for a production platform

· Independent learning and execution under a no-formal-training constraint